

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

Roll No. :S030	Name:Pradeep Rajbhar
Class :SYIT	Batch : 1
Date of Assignment :15/7/26	Date/Time of Submission :15/7/26

Queue Application:

- a. Write a program to: a. Implement a queue using an array.

Code:

dsprac4.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/dsprac4.py

File Edit Format Run Options Window Help

```
import array as arr

que = arr.array('i', [])

si = 4
|
print(que)

def enqueue(data):
    if(len(que) == si):
        print("Queue Overflow")
    else:
        que.append(data)
        print(que)

def dequeue():
    if(len(que) == 0):
        print("Queue Underflow")
    else:
        que.pop(0)
        print("Data removed: \n", que)

def peek():
    if(len(que) == 0):
        print("Queue is Empty")
    else:
        print("First element is: ", que[0])

def show():
    if(len(que) == 0):
        print("Queue is Empty")
    else:
        print("----- QUEUE ----- \n", que)

enqueue(2)
dequeue()
dequeue()
enqueue(3)
enqueue(4)
enqueue(5)
enqueue(6)
enqueue(7)

dequeue()
peek()
dequeue()
show()

print("Pradeep S030")
```

Output:

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944]
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/itlab/

array('i')
array('i', [2])
Data removed:
  array('i')
Queue Underflow
array('i', [3])
array('i', [3, 4])
array('i', [3, 4, 5])
array('i', [3, 4, 5, 6])
Queue Overflow
Data removed:
  array('i', [4, 5, 6])
First element is:  4
Data removed:
  array('i', [5, 6])
---- QUEUE ----
  array('i', [5, 6])
Pradeep S030
|
```

b. Simulate a simple queuing system (e.g., customer service queue).

Code:

```
File Edit Format Run Options Window Help

import time

que = [("C1",2), ("C2",5), ("C3",7), ("C4",0)]

print(" ----- Customer Order ----- ")
print(que)

while que:
    Order, timer = que.pop(0)
    print(f" Serving: {Order} {timer} sec")
    time.sleep(timer)

print(que)

print("Pradeep S030")
```

Output:

```
=====
----- Customer Order -----
[('C1', 2), ('C2', 5), ('C3', 7), ('C4', 0)]
Serving: C1 2 sec
Serving: C2 5 sec
Serving: C3 7 sec
Serving: C4 0 sec
[]
Pradeep S030
|
```